Developmental Stage Classification in Pigs

A Machine Learning Approach Using Transcriptomic Data

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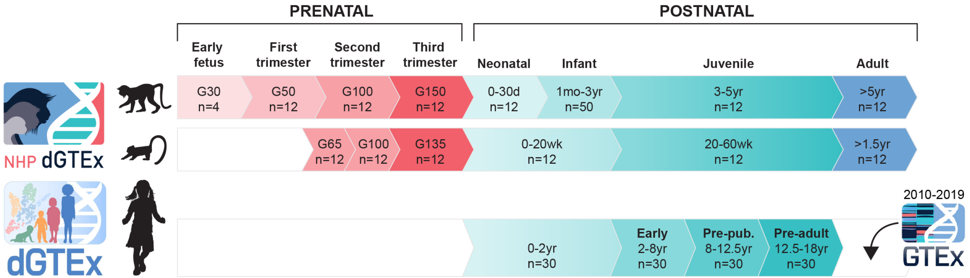
Background & Motivation

The Gap in Human Developmental Data



Comprehensive human developmental transcriptomics remains limited.

Coorens, T.H.H., Guillaumet-Adkins, A., Kovner, R. *et al.* The human and non-human primate developmental GTEx projects. *Nature* 637, 557–564 (2025). doi:10.1038/s41586-024-08244-9

The Challenge

Current Gaps

- No molecular standard to stage pigs; chronological age is unreliable.
- Tissue signals differ, so cross-tissue staging is inconsistent.
- Pig–human alignment is unclear; shared biomarkers are underused.

Our Goal

- **Gene Expression** $\rightarrow$ **Pig Age**: Define molecular developmental stages from transcriptomics.
- **Pig Age** $\rightarrow$ **Human Age**: Align pig stages to human timeline via conserved biomarkers.
- **Result**: A unified, cross-species molecular clock.

2 Data Overview

The PigGTEx Project

Context & Mission

- Part of the **FarmGTEx** consortium (Farm Animal Genotype-Tissue Expression).
- **Goal:** Build a comprehensive catalogue of genetic regulatory variants in pigs.
- Analyze **transcriptomic landscape** across diverse tissues and biological contexts.

Key Objectives

- **Genetic Mechanisms:** Link genetic variation to gene expression.
- **Biomedical Model:** Leverage pigs as models for human physiology and disease.
- **Sustainable Agriculture:** Inform breeding for health and efficiency.

A massive public resource: Integrating thousands of RNA-seq samples to map the pig regulatory genome.

Data Quality Challenges

Missing Data

- **52%** of samples missing age information
- **52%** of samples missing sex labels
- **13%** of samples with unknown tissue category

Age Format Inconsistencies Age

stored as text in various formats:

- "0 day", "1 day", "7 days"
- "1 week", "2 weeks", "4 weeks"
- "1 month", "3 months", "6 months"
- "1 year", "2 years"
- "unknown"

Challenge:
Harmonize to a unified numeric format

Data Processing Pipeline

1. Age Harmonization

- Parse text $\rightarrow$ days
- Map to 5 biological stages:

Infant	0–20 days
Early childhood	21–59 days
Pre-pubertal	60–149 days
Post-pubertal	150–365 days
Adult	¿365 days

2. Expression Preprocessing

- $\text{Log}_2(\text{TPM} + 1)$ transformation
- Z-score normalization

3. Quality Control

- Filter samples with missing stage
- Explicit "Unknown" sex encoding

4. Adaptive Classification

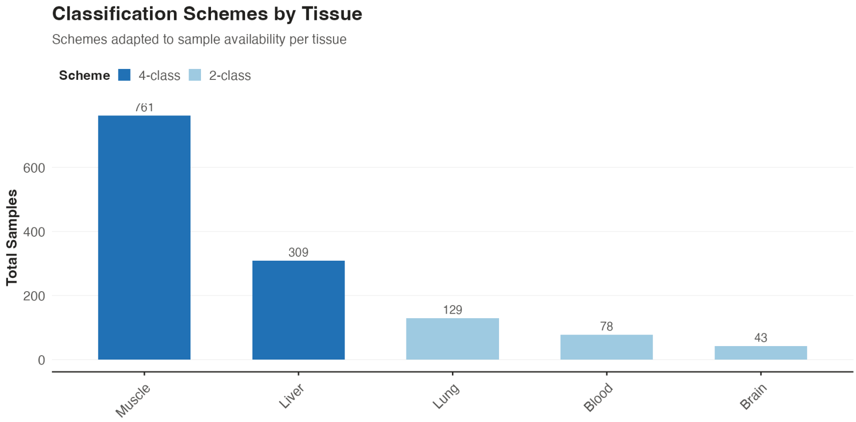
- 4-class: Muscle, Brain, Liver
- 3-class: Blood, Intestine
- Binary: Adipose, Testis

$\downarrow$

2,467 curated samples

across 8 major tissues

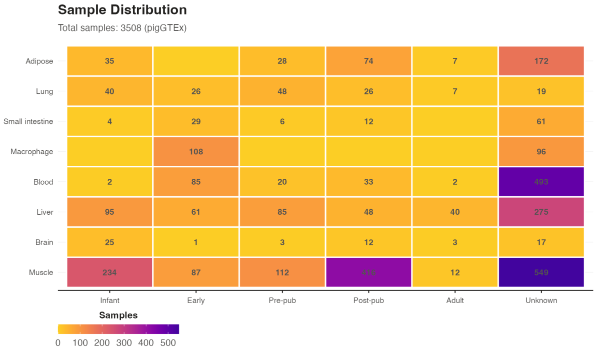
Classification Schemes



Dataset: pigGTEx

Scale & Scope

- **2,467 samples** across **8 tissues**.
- Comprehensive sequencing of:
    - Muscle, Brain, Liver, Blood
    - Intestine, Lung, Adipose, Testis



3 Methodology

Methodology Overview

1. Classification

- Multi-resolution schemes
- 4-class / 3-class / Binary

2. Machine Learning

- Ordinal LightGBM
- Binary reduction strategy

3. Validation

- Tissue-specific performance
- Cross-tissue robustness

The Machine Learning Problem

What is X? (Features)

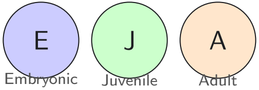
Gene expression values from RNA-seq

Gene1 = 5.2
Gene2 = 3.1
Gene3 = 8.7
... (20,000+ genes)

Each sample = one pig's gene expression profile

What is Y? (Label)

Developmental stage we want to predict

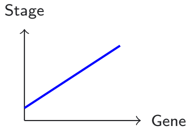


Goal: Gene expression $\rightarrow$ Predict stage

In simple terms: Given a pig's gene activity, which developmental stage is it in?

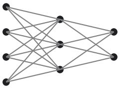
Why Machine Learning?

Traditional Regression



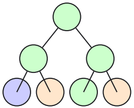
- Assumes linear relationships
- Genes interact non-linearly!

Deep Learning



- Needs **thousands** of samples
- We only have hundreds!

LightGBM ✓

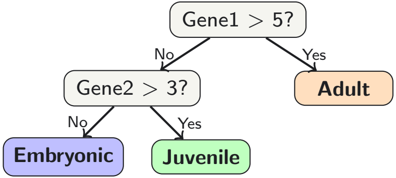


- Handles **non-linear** patterns
- Works with **small samples**

LightGBM: Best performance with limited sample sizes

How Does a Decision Tree Work?

Simple Example: 3 Genes to Predict Stage



How it works:

1. Start at the top
2. Ask yes/no questions about genes
3. Follow branches to decision

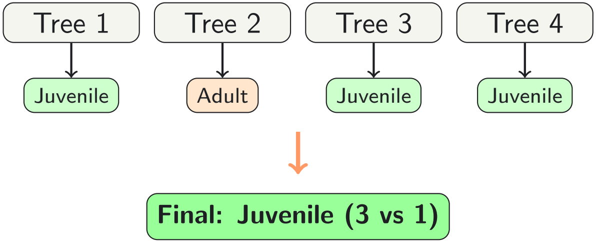
Example: Gene1=6.2, Gene2=2.1

- Gene1 $>$ 5? **Yes** $\rightarrow$ **Adult**

A single tree asks simple questions about genes to classify developmental stage.

From One Tree to Many Trees

One tree can make mistakes. Many trees vote together!



Why multiple trees?

- Each tree sees data differently
- Mistakes cancel out
- **Majority vote** is more reliable

More trees = More reliable predictions!

What Makes LightGBM Special: Boosting

Each new tree learns from previous mistakes

Tree 1 — Initial guess $\rightarrow$ Some errors

$\downarrow$ Focus on errors

Tree 2 — Fix Tree 1's mistakes $\rightarrow$ Fewer errors

$\downarrow$ Focus on remaining

Tree 3 — Refine further $\rightarrow$ Very few errors

...

$\downarrow$

Final Prediction — Weighted sum of all trees

Boosting: Trees work as a team, each fixing the others' mistakes

Why Ordinal Classification Matters

Developmental Stages Have Order

Embryonic → Juvenile → Adult

Natural ordering: E $<$ J $<$ A

Key insight:
Misclassifying Adult as Embryonic is **worse** than misclassifying as Juvenile!

Standard vs Ordinal Approach

Standard E J A All equal

Ordinal E — J — A Ordered!

Ordinal models:

- Respect stage ordering
- Penalize distant errors more
- Better calibrated probabilities

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Results

Performance Metrics Overview

Precision Proportion of positive identifications that were actually correct.

$$\text{Precision} = \frac{TP}{TP+FP}$$

Recall Proportion of actual positives identified correctly.

$$\text{Recall} = \frac{TP}{TP+FN}$$

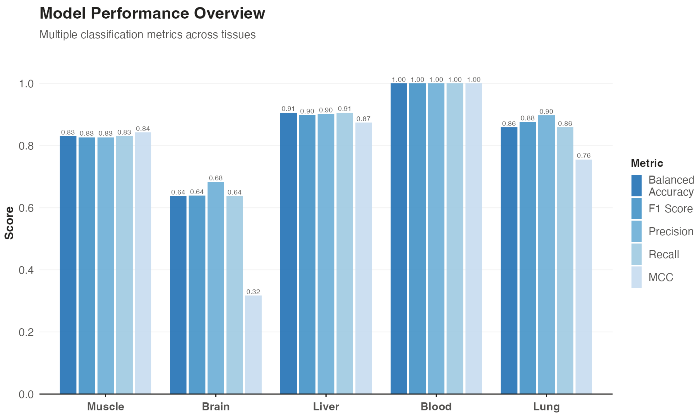
F1 Score Harmonic mean of Precision & Recall.

$$\text{F1} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

MCC (Matthews Correlation Coefficient) Balanced measure for imbalanced datasets.

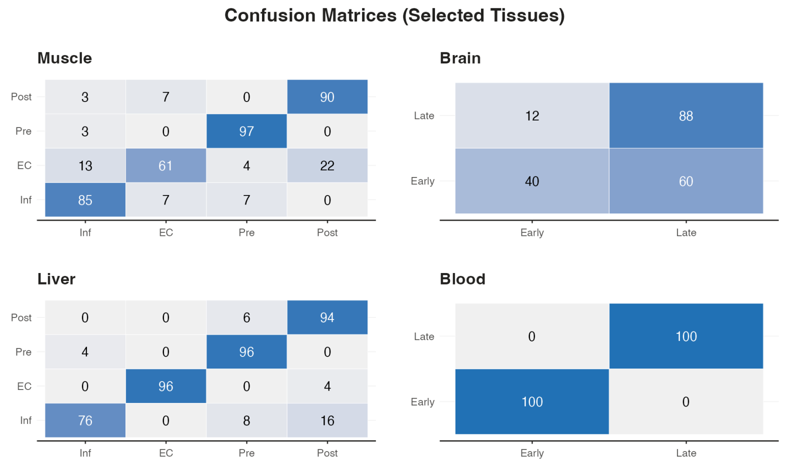
$$\text{MCC} = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP+FP)(TP+FN)(TN+FP)(TN+FN)}}$$

Model Performance

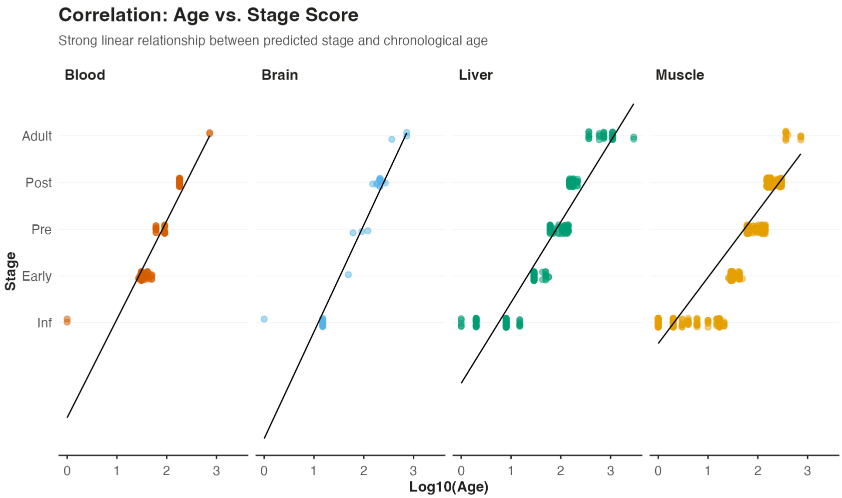


Strong performance across all metrics: Balanced Accuracy, F1, Precision, Recall, MCC

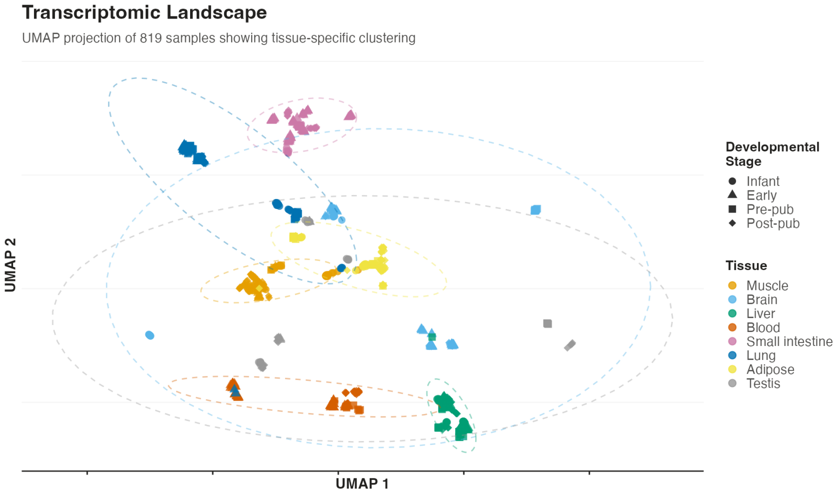
Confusion Matrices

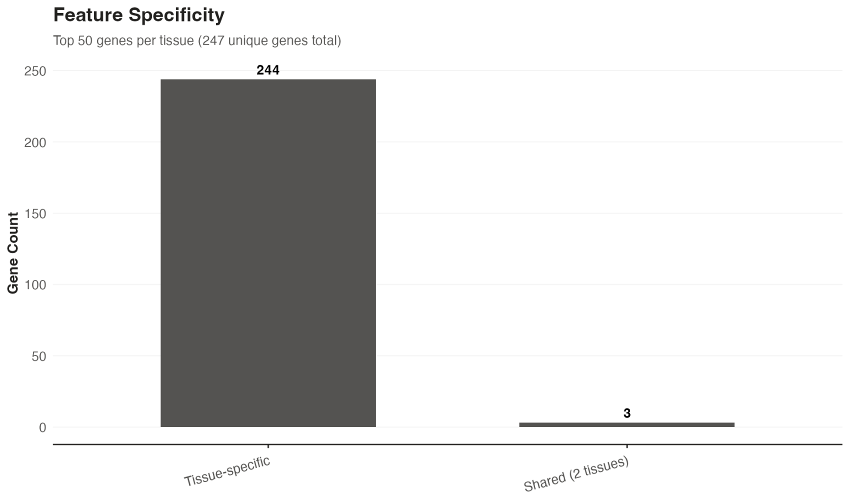


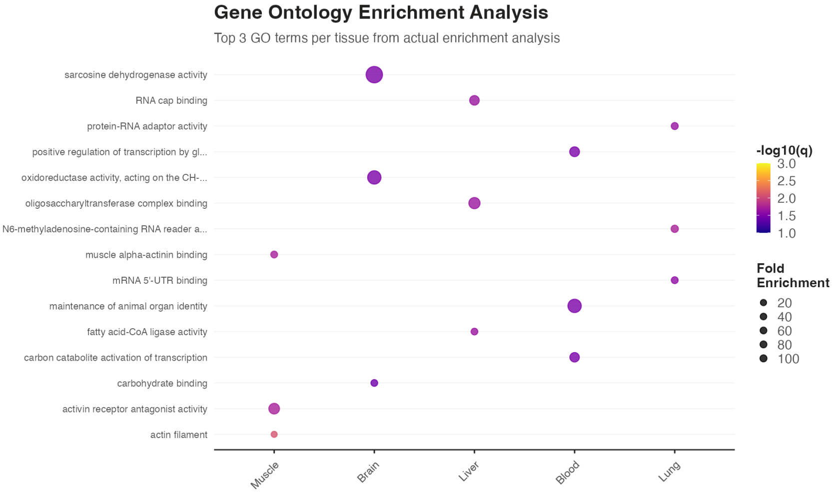
Age Correlation

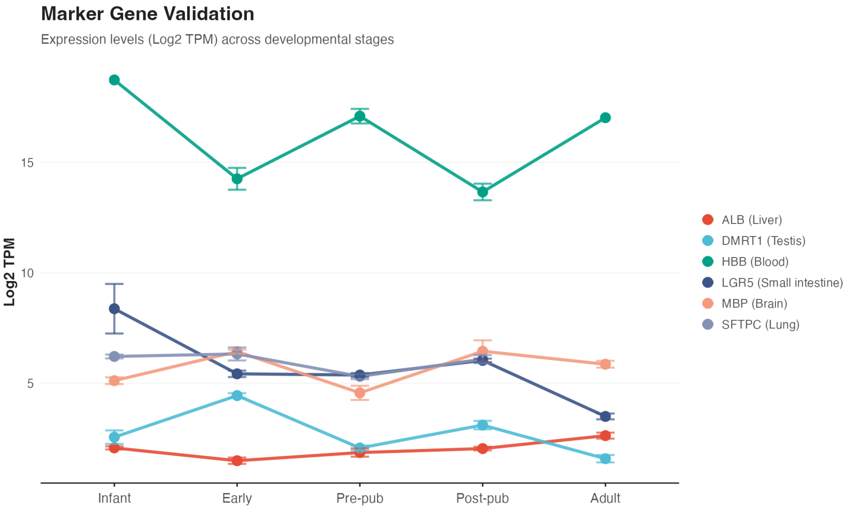


5 Current Model Issue


Transcriptomic Landscape
UMAP projection of 819 samples showing tissue-specific clustering
UMAP 2
UMAP 1
Developmental Stage
Infant
Early
Pre-pub
Post-pub
Tissue
Muscle
Brain
Liver
Blood
Small intestine
Lung
Adipose
Testis


Feature Specificity
Top 50 genes per tissue (247 unique genes total)
Gene Count
250
200
150
100
50
0
244
3
Tissue-specific
Shared (2 tissues)


Gene Ontology Enrichment Analysis
Top 3 GO terms per tissue from actual enrichment analysis
sarcosine dehydrogenase activity
RNA cap binding
protein-RNA adaptor activity
positive regulation of transcription by gl...
oxidoreductase activity, acting on the CH-...
oligosaccharyltransferase complex binding
N6-methyladenosine-containing RNA reader a...
muscle alpha-actinin binding
mRNA 5'-UTR binding
maintenance of animal organ identity
fatty acid-CoA ligase activity
carbon catabolite activation of transcription
carbohydrate binding
activin receptor antagonist activity
actin filament
Muscle
Brain
Liver
Blood
Lung
-log10(q)
3.0
2.5
2.0
1.5
1.0
Fold Enrichment
20
40
60
80
100


Marker Gene Validation
Expression levels (Log2 TPM) across developmental stages
Log2 TPM
15
10
5
Infant
Early
Pre-pub
Post-pub
Adult
ALB (Liver)
DMRT1 (Testis)
HBB (Blood)
LGR5 (Small intestine)
MBP (Brain)
SFTPC (Lung)

Conclusion

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Key Findings

- Accurate staging from transcriptomics
- Tissue-specific molecular clocks
- Conserved developmental programs

Next Steps

- Cross-species integration (Human/Mouse)
- Phenotypic data integration
- Disease modeling applications